Project Planning Phase

Project Planning Template (Product Backlog, Sprint Planning, Stories, Story points)

Date	8 March 2026
Team ID	Team 169
Project Name	Insurance Fraud Detection Using Machine Learning
Maximum Marks	8 Marks

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Use the below template to create product backlog and sprint schedule

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Data Collection	USN-1	As a developer, I collect insurance claim datasets for fraud detection	3	High	Ayushi
Sprint-1	Data Preprocessing	USN-2	Clean and preprocess the dataset by removing missing and duplicate values.	3	High	Himanshu
Sprint-2	Feature Engineering	USN-3	Select important features affecting insurance fraud prediction.	2	Medium	Kshitij

Sprint-2	Model Development	USN-4	Train machine learning models (Logistic Regression, Decision Tree, Random Forest).	4	High	Gaurav
Sprint-3	Model Evaluation	USN-5	Evaluate models using accuracy, precision, recall and F1-score.	3	High	Himanshu
Sprint-3	Hyperparameter Tuning	USN-6	Improve model performance using hyperparameter tuning.	2	Medium	Gaurav
Sprint-4	Result Visualization	USN-7	Create charts and confusion matrix for model performance.	2	Medium	Kshitij
Sprint-4	Documentation & Report	USN-8	Prepare final project report and presentation.	3	High	Ayushi

Project Tracker, Velocity & Burndown Chart: (4 Marks)

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	10	7 Days	02 Feb 2026	09 Feb 2026	10	09 Feb 2026
Sprint-2	10	7 Days	10 Feb 2026	17 Feb 2026	10	17 Feb 2026
Sprint-3	10	7 Days	18 Feb 2026	25 Feb 2026	10	25 Feb 2026

Sprint-4	10	7 Days	26 Feb 2026	08 Mar 2026	10	08 Mar 2026

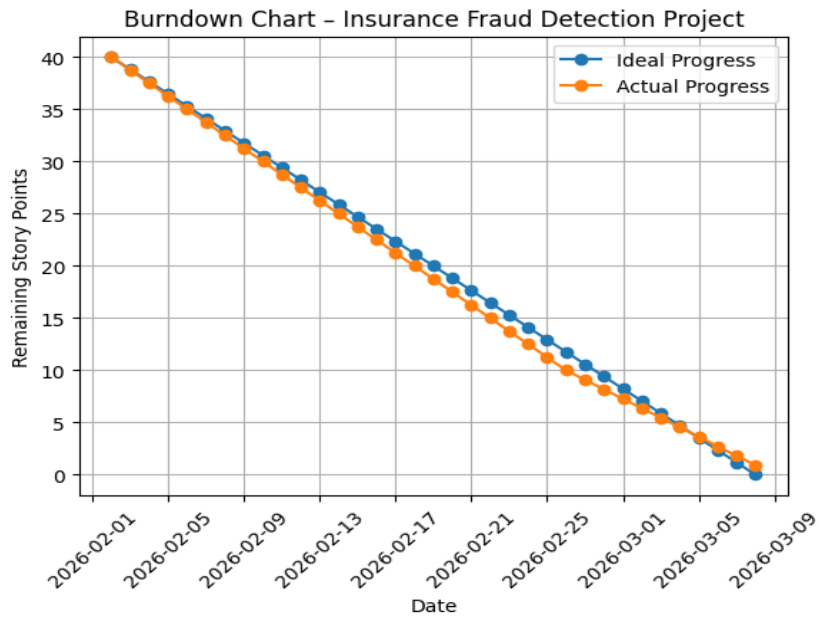
Velocity:
Imagine we have a 10-day sprint duration, and the velocity of the team is 20 (points per sprint). Let's calculate the team's average velocity (AV) per iteration unit (story points per day)

$$AV = \frac{\text{Story Points}}{\text{Sprint Duration}} = \frac{10}{7} = 1.42$$

* What this chart shows 📊

- **Ideal Progress Line: Shows how work should decrease evenly from 40 story points (2 Feb 2026) to 0 (8 Mar 2026).**
- **Actual Progress Line: Shows the real progress based on your sprint plan:**

- Sprint-1 → 10 points completed
- Sprint-2 → 10 points completed
- Sprint-3 → 10 points completed
- Sprint-4 → 10 points completed



Burndown Chart:

The burndown chart represents the remaining work in the project over time. The project started with **40 total story points on 2 February 2026**. The **ideal line** shows the expected linear progress if the team completed work at a constant rate. The **actual progress line** represents the real work completed during each sprint. By **8 March 2026**, all story points were completed, indicating that the project was successfully finished within the planned schedule.

https://www.visual-paradigm.com/scrum/scrum-burndown-chart/

https://www.atlassian.com/agile/tutorials/burndown-charts

Reference:

https://www.atlassian.com/agile/project-management

https://www.atlassian.com/agile/tutorials/how-to-do-scrum-with-jira-software

https://www.atlassian.com/agile/tutorials/epics

https://www.atlassian.com/agile/tutorials/sprints

https://www.atlassian.com/agile/project-management/estimation

https://www.atlassian.com/agile/tutorials/burndown-charts